A Unified Low-Mode Field Framework for Critical Mode Mixing and Family Structure

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Abstract

We propose a unified low-mode field framework describing critical mode mixing, slow-mode formation, and family structure generation across different physical systems. The theory is formulated using a mother field $\mathbb U(x,\sigma)$ defined on an extended manifold $\mathcal M_4 \times \Sigma$, where Σ is a compact internal geometry.

By performing internal eigenmode decomposition and low-mode truncation, the system dynamics reduce to an effective two-mode structure described by parameters (A,B,C,Γ). These parameters are further compressed into dimensionless variables (δ,c,γ), generating a universal kernel

$$\mathcal E(\delta,c,\gamma)=(\Pi,r,\theta,\bar\tau)$$

that governs mixing strength, spectral splitting, mixing angle, and critical slowdown.

The framework predicts a universal stability condition

$$\delta^2+c^2 < 1$$

with a critical boundary $\delta^2+c^2 = 1$ corresponding to slow-mode formation.

The theory also naturally generates three family states through internal eigenmode windows. Numerical scans demonstrate the emergence of a universal phase diagram describing weak mixing, hybridization, and slow-mode regions.

This work provides a unified low-energy description of mode mixing phenomena and suggests a cross-system language for critical dynamics.

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1 Introduction

Mode mixing and critical slowdown appear in many physical systems including • photonic crystals • phonon dispersions • magnon hybridization • quantum level repulsion.

Despite their different physical origins, these systems often exhibit remarkably similar spectral behaviors near avoided crossings or band edges.

Existing theoretical descriptions are usually system-specific and lack a unified structural explanation.

In this work we introduce a low-mode field framework capable of describing these phenomena using a common mathematical structure.

The key idea is that many complex systems can be reduced to a small number of dominant modes. When two modes dominate the dynamics, the system effectively becomes a two-mode interaction problem whose structure can be expressed using a universal dimensionless kernel.

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2 Unified Field Framework

We consider a mother field

$$\mathbb U(x,\sigma)$$

defined on the extended manifold

$$\mathcal M_4 \times \Sigma$$

where • x represents spacetime coordinates • σ parameterizes an internal compact geometry.

2.1 Unified Action

The theory is defined by

$$S= \int_{\mathcal M_4 \times \Sigma} d^4x,d\mu_\Sigma \sqrt{|g|} \left(\mathcal L_G + \mathcal L_U + \mathcal L_\Psi + \mathcal L_Y + \mathcal L_{proj} \right)$$

where

Term Meaning $\mathcal L_G$ gravitational background $\mathcal L_U$ mother-field dynamics $\mathcal L_\Psi$ fermionic sector $\mathcal L_Y$ Yukawa coupling $\mathcal L_{proj}$ projection correction

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2.2 Mother-Field Dynamics

$$\mathcal L_U = (D_M\mathbb U)^\dagger D^M\mathbb U -\zeta \mathbb U^\dagger \Delta_\Sigma \mathbb U -\mathbb V(\mathbb U)$$

with potential

$$\mathbb V(\mathbb U)$$

$$\mu^2|\mathbb U|^2 + \lambda|\mathbb U|^4$$

Variation gives

$$D^M D_M \mathbb U -\zeta \Delta_\Sigma \mathbb U + \frac{\partial \mathbb V}{\partial \mathbb U}$$

$$\mathcal J_{proj}$$

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3 Internal Geometry

To define the eigenmode structure we specify the internal space

$$\Sigma = S^2$$

the unit two-sphere.

The Laplacian eigenproblem is

$$\Delta_\Sigma Y_{\ell m}$$

$$-\ell(\ell+1)Y_{\ell m}$$

giving eigenvalues

$$\Lambda_\ell = \ell(\ell+1)$$

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4 Mode Decomposition
The mother field expands as

$\mathbb U(x,\sigma)$

$\sum_{\ell,m} Y_{\ell m}(\sigma)\phi_{\ell m}(x)$
where
$\ell=0,1,2,\dots$
Near critical regions only the lowest modes contribute significantly.

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5 Low-Mode Reduction
Keeping two dominant modes yields
$M= \begin{pmatrix} 2A & C\\ C & 2B \end{pmatrix}$
Eigenvalues

$\lambda_\pm$

$(A+B) \pm \sqrt{(A-B)^2+C^2}$

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6 Dimensionless Universal Kernel
Define
$\delta=\frac{A-B}{A+B}$
$c=\frac{C}{A+B}$
$\gamma=\frac{\Gamma}{A+B}$
Let
$\rho=\sqrt{\delta^2+c^2}$

Kernel observables
Mixing strength
$\Pi=\frac{c^2}{1-\delta^2}$
Spectral ratio
$r=\frac{1+\rho}{1-\rho}$
Mixing angle
$\theta=\frac12\arctan\frac{c}{\delta}$
Slow-mode time scale
$\bar\tau=\frac{1+\Gamma}{1-\rho}$

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7 Critical Structure
Stability requires
$\lambda_- >0$
leading to
$\boxed{\delta^2+c^2<1}$
Critical boundary

$$\delta^2+c^2=1$$

Near this boundary

$$\tau\sim\frac1{1-\rho}$$

indicating critical slowdown.

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8 Universal Phase Diagram

Scanning (δ,c) produces a universal phase diagram.

Region Condition Physics Weak mixing $\rho<0.3$ modes independent Hybridization $0.3<\rho<0.7$ avoided crossing Slow-mode $\rho\to1$ critical slowdown

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9 Numerical Scan Example

Example parameters

$A = 1.0$ $B = 0.6$ $C = 0.7$ $\Gamma = 0.02$

Compute

$$S=A+B=1.6$$

$$\delta=0.25$$

$$c=0.4375$$

$$\rho=0.503$$

Slow-mode time scale

$$\tau \approx 2.0$$

indicating moderate hybridization.

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10 Physical Applications

The universal kernel applies across multiple systems.

System Interpretation Photonic crystals band-edge slow light Phonon dispersions soft-mode formation Magnons mode hybridization Quantum circuits level repulsion

Each system maps its parameters to

$$(A,B,C,\Gamma)$$

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11 Family Structure

Internal eigenvalues satisfy

$$\Lambda_\ell=\ell(\ell+1)$$

A low-mode window

$$\Lambda_\ell<\Lambda_c$$

with

$$\Lambda_c\approx6$$

gives

$$\ell=0,1,2$$

producing three families.

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12 Testable Prediction

Mode overlap integrals

$$Y_{ij}\sim\int_\Sigma u_i^\dagger \mathcal O u_j$$

predict hierarchical suppression

$|Y_{02}| \ll |Y_{01}|,|Y_{12}|$
indicating weaker mixing between distant families.

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13 Discussion

The proposed framework provides a unified language for mode mixing phenomena.
Limitations include • lack of detailed experimental parameter extraction • need for numerical simulations • higher-mode corrections.
Future work will explore cross-system validation.

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14 Conclusion

We introduced a unified low-mode field framework describing critical mode mixing and slow-mode formation across physical systems.
The theory reduces complex dynamics to a dimensionless kernel
$\mathcal E(\delta,c,\gamma)$
revealing a universal critical structure
$\delta^2+c^2=1$
and naturally generating three family states through internal eigenmode windows.

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